

Jan Kazimierczak

Engineering Science Student | University of Toronto

jankazimierczak.github.io/Website/ | [GitHub](#) | [LinkedIn](#)

Languages: Polish (native) | Spanish (native) | English (C1)

Second-year student with an intended aerospace specialization and project experience in structural simulation, experimental aerodynamics, machine vision and physical prototyping.

Education

University of Toronto

Expected May 2030

Bachelor of Applied Science in Engineering Science

Cumulative GPA: 3.58/4.00 | Intended specialization: Aerospace Engineering

Expected graduation includes Professional Experience Year (PEY) co-op.

Relevant completed coursework: Structures and Materials; Calculus I and II; Engineering Mathematics and Computation; Electric Circuits; Computer Programming (C and Python); Algorithms and Data Structures; Engineering Design (Praxis I and II).

In progress (Fall 2026): AER210 Vector Calculus and Fluid Mechanics; CHE260 Thermodynamics and Heat Transfer; MAT292 Ordinary Differential Equations; ECE253 Digital and Computer Systems; PHY293 Waves and Modern Physics.

Planned coursework (Winter 2027): PHY294 Quantum and Thermal Physics; ESC204 Praxis III (mechatronic design); ECE259 Electromagnetism; MIE286 Probability and Statistics.

Student Design Team

University of Toronto Aerospace Team

Jan 2026 - Present

Liquid Propulsion Member

Technical Skills

Programming and analysis: Python, C, MATLAB, JavaScript, LaTeX

CAD and simulation: Fusion 360, Onshape, ANSYS Fluent

Prototyping and testing: 3D printing, Raspberry Pi, sensor-data analysis, structural modelling, physical testing and technical reporting

Digital systems (current coursework): SystemVerilog, ModelSim, FPGAs and digital logic

Engineering Projects

Grid Fin Aerodynamics and Wind Tunnel Testing

Sep 2023 - Apr 2024

European Schools Science Symposium | Three-person team

- Designed and built a custom 3 m wind tunnel, then wired and instrumented a load-cell rig using an HX711 and Arduino Uno for a model-scale grid-fin study.
- Designed 18 grid-fin variants in Fusion 360 and 3D printed them, comparing two outer outlines, three hole shapes and three nominal hole sizes.
- Captured more than 180,000 sensor readings through a Python-to-CSV pipeline; analyzed the data in R using visualizations, ANOVA and Tukey HSD comparisons.
- Co-authored the team report. The simple-outline fin with 6 mm square holes produced the highest measured load-cell response within the tested design matrix.

Matboard Bridge Structural Optimization

Nov 2025

CIV102 | University of Toronto | Three-person team

- Wrote Python simulations of a moving train load to compare bridge geometries and predicted failure loads; checked selected outputs with three hand calculations.
- Evaluated four geometry iterations, increasing the model-predicted minimum failure load from 240 N to 869.5 N, a 3.6-fold increase.
- Fabricated and load-tested the 1.265 m bridge. Post-test analysis identified failure at a splice/glue joint absent from the analytical model.

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Engineering Projects Continued

Machine Vision for an Automated Screw Sorter

Jan 2026 - Apr 2026

Praxis II | University of Toronto

- Generated synthetic training images, trained and tested a convolutional neural network (CNN), and integrated camera-based inference on a Raspberry Pi to classify screw-head shapes.
- The source dataset contained 153 original images across three head types before augmentation. Evaluated classification at approximately 70% in-session and 80% on a separate 34-screw manual test.
- Compared the results with the 97.5% accuracy requirement and documented the shortfall. Testing covered the vision subsystem; the complete sorting system was not tested as an integrated unit.

BikePack Buddy Foldable Basket Extension

Sep 2025 - Dec 2025

Praxis I | University of Toronto

- Generated 12 carrier concepts across five bicycle mounting locations for a portable backpack carrier compatible with Toronto Bike Share bicycles.
- Helped narrow 50 team ideas to five candidates using a morphological chart, SCAMPER, a measurement matrix and four Pugh charts.
- Processed accelerometer and gyroscope data in MATLAB for four mounting positions at 5 ± 1 km/h. One proxy trial per position supported the team's selection of a foldable front-basket extension.

Software Project

Student Productivity and AI Research Platform

Personal project | Python, FastAPI, JavaScript, Docker, Raspberry Pi, Tailscale

- Built a full-stack platform to improve personal productivity by centralizing Kanban tasks, schedules, grade analytics, documents and application tracking across eight academic terms.
- Developed companion macOS and iPhone apps connected through Tailscale and scoped task APIs, enabling task creation, deadline updates and status changes across devices.
- Integrated OpenClaw for AI-assisted opportunity research and personalized outreach drafts, with content validation and single-use approval tokens tied to exact email and attachment content.
- Added atomic writes, crash-recovery journaling and automated tests covering authentication, concurrent updates, mobile APIs and outreach workflows.

Leadership and Community Experience

IT Officer

Sep 2024 - Jul 2025

Pupils' Committee | European School of Alicante | Alicante, Spain

- Maintained the committee's official website and delivered IT projects requested by the committee president.

Community Kitchen Volunteer

Jun 2024 - Jul 2024

Alicante Gastronómica Solidaria | Alicante, Spain

- Prepared meals in a community kitchen serving people experiencing homelessness and food insecurity.

Youth Basketball Coach

Jun 2023 - Jul 2023

Fundación Lucentum Alicante | Alicante, Spain

- Designed drills and coached children at a summer camp, working alongside professional coaches and independently.